

Aaron Xiong

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EDUCATION

University of Michigan

Ann Arbor, MI, USA

Bachelor of Science in Engineering in Computer Science, Minor in Electrical Engineering

May. 2027

Cumulative GPA: 3.972/4 (Dean's Honors List)

(Expected)

Courses: Data Structures and Algorithms, Computer Organization, Embedded Systems, Computer Security, Linear Algebra.

SKILLS

- **Programming Languages:** C/C++, Python, Swift, HTML/CSS, JavaScript, MATLAB, Verilog, ARMv7, NIOS II, Arduino, SQL
- **Libraries & Frameworks:** ReactJS, Flask, Tailwind CSS, Bootstrap, PyTorch, NumPy, PySerial, Matplotlib, GTK, Libcurl
- **Tools:** Linux Ubuntu, Git, Docker, Oracle, VMWare

WORK EXPERIENCE

NLP Research Assistant at LAUNCH Lab

Mar. 2025 – Present

- Developed a comprehensive Open-source LLM benchmark on Olympiad-Level Code Generation with **430+** Informatics Olympiad problems from **60+** competitions between 2023 and 2025, featuring hidden test suites and human contestant rankings.
- Extracted, restructured, and processed task files for each competition and built a contestant database by locating profiles with Serper and parsing Codeforces rating histories, using Google Gemini AI for profile matching based on name, country and text.
- Evaluated performances of **20+** LLMs against human contestants and systematically computed Elo ratings for each LLM model to benchmark code generation capabilities based on test suite results.

Full-stack Developer for U-M School of Nursing

Jan. 2025 – Present

- Designed a HIPAA-compliant, mobile-friendly, Full-stack web platform with a team of seven that aims to provide Nurse Health Coaching, Educational Content, and Personalized Community Resources for pregnant people and infant caregivers.
- Integrated Zoom and Qualtrics for nurse coaching and patient assessments, Google Maps for geospatial analysis, Oracle Database for patient data storage, and U-M's text-based generative AI tool Maizey for summarization functionalities.
- Engineered and optimized prompts for Maizey to accurately summarize meeting notes from nurse-patient Zoom sessions, generate personalized care plans for clients, and extract key needs from health surveys while mitigating hallucinations to ensure safety.

PROJECT EXPERIENCE

LockeDown (Computer Vision, LLM, Python)

Sep. 2025

- Developed a Full-stack real-time posture and productivity monitoring system integrating React, FastAPI, and SQLite.
- Leveraged YOLOv8m-pose and OpenCV to build a Linux program that detects slouching and delivers audio alerts to the user, integrating MemryX accelerators to optimize video inference and adopting a multi-threaded architecture for seamless performance.
- Monitored application sessions and classified productivity with Gemini, caching past sessions with SQLite3 to reduce token usage.
- Created a gamified scoring system with leaderboards, historical analytics, and real-time dashboards to encourage user engagement.

Relational Query Engine (C++)

Mar. 2025

- Implemented a lightweight relational database engine in C++ that supports SQL operations for structured data management.
- Integrated the Field class to store and process heterogeneous data types, enabling flexible storage, lookups, and comparisons.
- Enhanced query performances by **25-48%** by integrating hash-based and index-based indexing, optimized data structures for memory efficiency, and refined the system for deployment in realistic resource-constrained environments.

Fake News Classifier (C++, Machine Learning)

Sep. 2024 – Dec. 2024

- Designed an ML algorithm in C++ to identify misinformation in news articles using a training dataset of **12,000+** articles.
- Implemented a Naive Bayes Classifier to predict article authenticity based on linguistic patterns and created a preprocessing pipeline with text normalization, N-gram modeling, and Word Stemming, improving model accuracy **from 62% to 83.8%**.
- Delivered experimental results to stakeholders through a technical presentation, explaining methodology and evaluating trade-offs.

Evie (C++, GIS, Algorithms)

Jan. 2024 – Apr. 2024

- Developed a Geographic Information System in C++ for electric vehicle users across multiple regions with a team of three.
- Extracted real-world map data from StreetsDatabase and OpenStreetMapDatabase, rendered visuals using EZGL and GTK, integrated live weather data with Libcurl, designed a user-friendly interface using Glade, and managed team data with Git.
- Implemented A* Search for efficient navigation functionalities in the application and a TSP algorithm to optimize delivery routes, reducing the average path length by about **20%** relative to the initial implementation and ranking **top 15 out of 91** submissions.
- Led collaborative coding efforts by strategically dividing and distributing tasks and tracking team progress with Agile Roadmaps.